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## **Geomechanical Evaluation of Shale Caprock Integrity for Effective Carbon Storage**

N. N. S. Suhaidi, N. Ridzuan, and N. Ismail, Faculty of Chemical and Process Engineering Technology, Universiti Malaysia Pahang Al-Sultan Abdullah, Gambang, Pahang, Malaysia

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### **Abstract**

This study analyze the geomechanical behavior of shale cap rocks during CO<sub>2</sub> injection for Carbon, Capture and Storage (CCS) and specifically, the mechanisms of rock failure. It compares shale properties of two major CO<sub>2</sub> storage sites of the United States (Shale 1 & Shale 2) with a local Malaysian shale formation (Shale 3). The findings are intended to improve the understanding of caprock security and risk reduction in order to achieve safer storage methods. The shale samples (Shale 1, Shale 2, and Shale 3) were characterized for their mineralogical and morphology features by FESEM and EDX analysis. TGA were conducted to study thermal stability and triaxial compression tests were carried to evaluate the mechanical properties. A reservoir model was developed in tNavigator reservoir simulator to simulate CO<sub>2</sub> injection behavior in the local Malaysian shale (Shale 3), and results were compared with literature data from two U.S. formations (Shale 1 & 2) with successful CCS project. Mineralogical investigation showed quartz as the major mineral in all the shale samples while the amount of clay and organic matter varied. Regarding the FESEM imaging, Shale 1 and Shale 2 were of lower porosity which was due to higher content of quartz in these shales. EDX analysis identified elevated calcium and oxygen concentrations in Shale 1 and Shale 2, suggesting differing geochemical properties from Shale 3. The TGA showed thermal decomposition of all samples at approximately 650°C, with Shale 3 exhibiting superior thermal stability which could help to maintain structural integrity under thermal stresses pertaining to CCS. Mechanical properties of Shale 1 and Shale 2 were greater than those of Shale 3, as shown by the testing results. Reservoir simulation indicated that despite its different mineralogical properties, Shale 3 had comparable CO<sub>2</sub> storage capacity to the other two samples. These findings suggest that rock failure risks are influenced by both mechanical and geochemical interactions. This paper offers new insights into shale caprock failure mechanisms under CO<sub>2</sub> injection, providing crucial data for optimizing CCS site selection and enhancing storage security, particularly in Malaysia. By combining experimental analysis with advanced reservoir simulation, the study contributes to the development of safer and more efficient CCS practices for long-term CO<sub>2</sub> storage.

**Keywords:** CO<sub>2</sub> Storage, Shale Caprock Integrity, Mohr Coulomb, tNavigator, Reservoir Simulation

## Introduction

One of the promising pathways to reduce the concentration of greenhouse gases (GHGs) in the atmosphere is to store CO<sub>2</sub> into deep underground geological formations, known as carbon capture and storage (CCS) [1]. One of the crucial structures for the safety of a CCS project, among others, is the caprock which seals the injected CO<sub>2</sub> from migrating upwards [2], [3]. The caprocks are commonly made of shale, a low-permeability fine-grained sedimentary rock [4]. It is important to maintain the integrity of the caprock for the temporal confinement of CO<sub>2</sub> in the storage site in order to avoid leaking into other zones [5]. The objective of this analysis is to explore the stability of the overlying shale caprock under varying geomechanical and geochemical conditions. The aim is to characterize mineralogically, morphologically and physically, caprock shales from various formations. Secondly, the thermal behavior of such shale samples has been investigated, which is of relevance for CO<sub>2</sub> storage. Third, to simulate CO<sub>2</sub> injection for sealing effectiveness and failure assessment of the caprock. Three shale cores, named Shale 1, Shale 2, and Shale 3, are studied. Most of the studies about CO<sub>2</sub> storage have leaned overperformance, reservoir and injectivities behavior and the caprock, with limited study on the mechanical response of caprocks to CO<sub>2</sub> exposure, especially shale [6], [7]. This work is novel in its holistic study that combines laboratory mechanical testing results, mineralogical characterization, and reservoir simulation to evaluate the geomechanical well-being of shale caprocks under CO<sub>2</sub> injection.

Apart from such comparisons of local shale with well-known formations from U.S, the present study fills the existing knowledge gap. Although a number of studies have been carried out in Northern Europe and North America, properties and performance of Malaysian caprocks are less studied [8], [9], [10], [11]. With Malaysia's distinctive geological conditions, and due to its recent participation in CCS projects, it is important to assess the performance of the local shale formations for a safe and sustainable long-term CO<sub>2</sub> storage.

## Methodology

This work applies a coupled approach to investigate the geomechanical performance and sealing integrity of shale caprock during CO<sub>2</sub> injection. The methodology integrates laboratory mineralogical and mechanical testing, petrophysical analysis, and numerical modeling in tNavigator. The methodology is designed to evaluate mechanical and geochemical stability under stress conditions typical of CO<sub>2</sub> storage settings.

### Sample Selection and Data Source

Two different lithologies were studied, shale as caprock, and sandstone as reservoir. Three samples were studied Shale 1 and Shale 2 extracted from well characterized CO<sub>2</sub> storage formation in USA, and National Energy Technology Laboratory (NETL) provides the measurement data. Shale 3 and sandstone unit sample were originating from a local Malaysian formation and the reservoir data was obtained from PETRONAS. The ability to measure local formations based on established international standards is made possible by this generic framework.

### Mineralogical and Morphological Characterization

Mineralogical and morphological analysis was carried out using Field Emission Scanning Electron Microscopy (FESEM) and Energy Dispersive X-ray Spectroscopy (EDX). Bulk mineralogy compositions were analyzed using X-ray fluorescence (XRF) and to differentiate clay-rich shale from quartz-rich sandstone. These experiments showed that there are differences with regard to porosity, grain and mineral composition, which have a direct impact on the geomechanical behavior and the CO<sub>2</sub> interaction capacity.

### Thermal Stability Analysis

Thermal stability of the shale samples was assessed using Thermogravimetric Analysis (TGA) at a low and high heating rates of 10 °C/min and 20 °C/min. The decomposition onset temperature and residual mass were recorded to evaluate the thermal resilience of each shale sample under subsurface conditions. Shale 3 showed the highest thermal stability, suggesting it may offer higher structural integrity during CO<sub>2</sub> injection.

### Mechanical Testing

Triaxial compression testing was conducted on both shale and sandstone samples under a range of confining pressures. Cylindrical plugs were prepared in accordance with ASTM D7012 standard. The tests provided essential mechanical parameters including Young's Modulus, Poisson's Ratio, cohesion, and internal friction angle. These values were used as direct input for geomechanical simulation and analysis.

### Petrophysical Analysis

The petrophysical model was developed using tNavigator's Model Designer module. A simple box grid was created based on the thickness and lateral extent of the reservoir, referencing well log data to guide vertical layering. A CO<sub>2</sub> injection well (INJ\_1) was created using cross-sectional well placement to enable later simulation stages. Well log data were imported in LAS format, and well names were matched precisely to ensure compatibility.

Blocked well logs were created to map petrophysical properties into the grid, including porosity (PORO), water saturation (SW), and permeability (PERM). Gamma ray (GR) log values were analyzed to differentiate lithologies, and to distinguish shale and sandstone zones. These were applied during facies and property assignment, allowing rock properties to be distributed appropriately across the model.

Shale volume ( $V_{sh}$ ) and effective porosity were calculated from the blocked logs, and saturation values were derived using a combination of default log-based correlations and manual adjustments based on core-informed ranges. The resulting property models were used not only to initialize the simulation but also to define spatially variable inputs for the geomechanical module.

### Numerical Geomechanical Simulation

Numerical simulations were carried out in tNavigator to evaluate the caprock's mechanical response during CO<sub>2</sub> injection. The model domain included overburden and underburden layers in addition to the reservoir. Rock mechanical properties obtained from laboratory testing were mapped to the respective rock types based on facies.

Initial and boundary conditions were defined using regional stress trends, incorporating vertical and horizontal stress magnitudes and pore pressure gradients. The lower boundary was fixed in all directions to simulate subsurface confinement. The coupled simulation approach was implemented under fully implicit conditions. Injection schedules, well controls, and simulation periods were defined based on scenario planning. The model tracked changes in pressure distribution, stress fields, and displacement vectors over the injection timesteps.

Mechanical failure risk was assessed using the Mohr-Coulomb failure criterion. Simulated outputs were analyzed for indicators of plastic deformation, tensile fracturing, and shear slip zones. Special emphasis was placed on the containment performance of the shale caprock and potential leakage pathways across lithological interfaces. The outcomes of the simulation provided critical insights into the geomechanical integrity and feasibility of the reservoir-caprock system for long-term CO<sub>2</sub> storage.

## Results and Discussion

### Morphological Characterization via FESEM

Field Emission Scanning Electron Microscopy (FESEM) analysis at 15,000X magnification revealed distinct morphological differences among the three shale samples (Fig. 1).

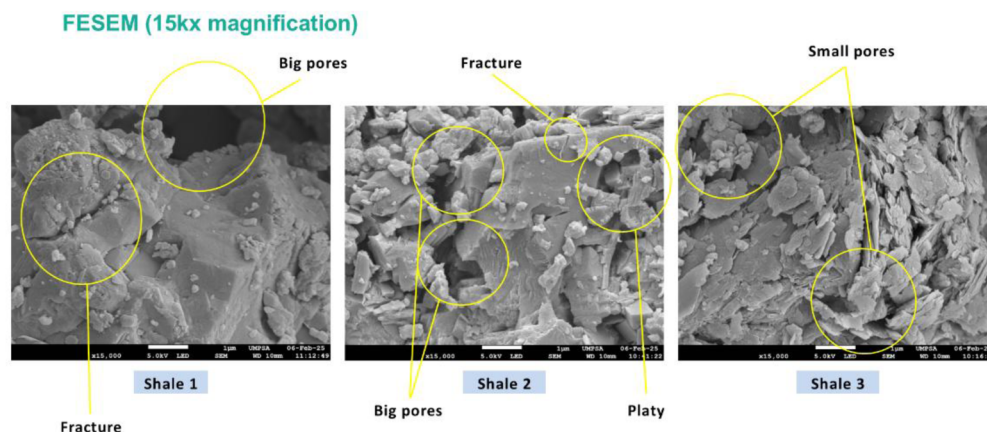


Figure 1—Field Emission Scanning Electron Microscope (FESEM) images of Shale 1, Shale 2, and Shale 3 at 15,000× magnification.

All samples displayed clear pore structures, however the size and nature of the pores were quite different, and thus so would the mechanical properties. The largest average pore size was found in Shale 1; the greatest total numbers of pores found in Shale 2. These properties indicate poor mechanical strength as well as reduced sealing ability in both materials. In addition, visible fractures were observed in Shale 1 and Shale 2, which can reduce their feasibility as reliable caprock rocks for CO<sub>2</sub> storage. Shale 3, however, had smaller, more evenly distributed pores, which are commonly considered to promote sealing capacity and caprock integrity.

Additionally, Shale 3 displayed a prominent platy morphology, with minor traces of similar structures identified in Shale 2. Previous studies have correlated platy features with high clay content in shale, implying that Shale 3 may possess a more clay-rich composition. This structural characteristic not only supports better self-sealing behavior but also contributes to greater geomechanical resilience under stress. Collectively, these observations suggest that Shale 3 is morphologically more favorable as a caprock material compared to Shale 1 and Shale 2.

### Energy Dispersive X-ray Spectroscopy (EDS)

Fig. 2 presents the Energy Dispersive X-ray Spectroscopy (EDS) results provide the bar chart of elemental distribution across shale sample 1, 2, and 3.

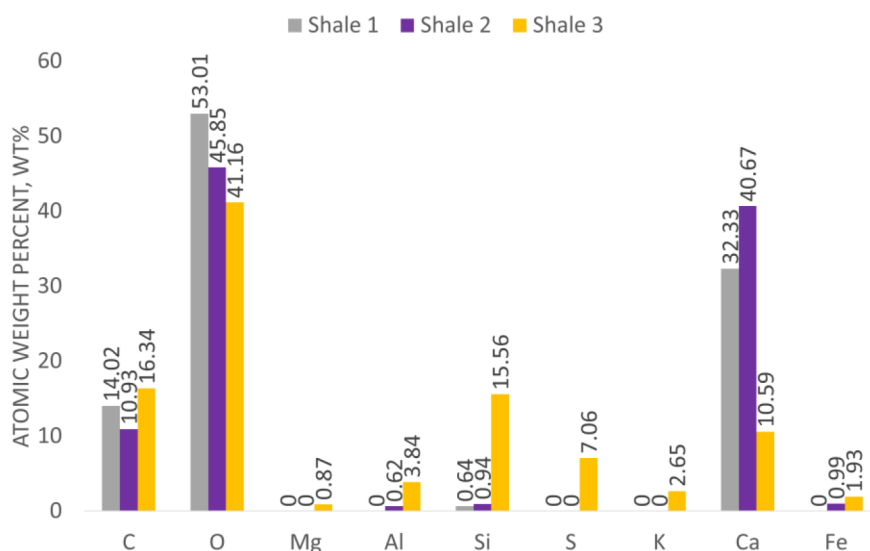


Figure 2—Elemental composition of Shale 1, Shale 2, and Shale 3 obtained via Energy Dispersive X-ray Spectroscopy (EDS).

Carbon (C) and oxygen (O) were the most abundant elements, consistent with the presence of carbonate minerals and organic matter. Shale 1 exhibited the highest carbon content (43.92 wt%), followed by Shale 2 (33.17 wt%) and Shale 3 (28.04 wt%). A similar trend was observed for oxygen, with Shale 1 containing 53.04 wt%, Shale 2 with 48.76 wt%, and Shale 3 at 40.62 wt%. This indicates a higher concentration of carbonate constituents, particularly calcite ( $\text{CaCO}_3$ ), in Shale 1 and Shale 2.

Silicon (Si), primarily associated with quartz ( $\text{SiO}_2$ ), was significantly higher in Shale 3 (15.56 wt%) compared to Shale 2 (6.84 wt%) and Shale 1 (0.64 wt%). This suggests that Shale 3 has a more siliceous composition, which aligns with XRF findings indicating higher sandstone content in this sample. The elevated quartz content in Shale 3 is likely to enhance its mechanical strength, a key factor in caprock integrity.

Aluminum (Al) and potassium (K), elements indicative of clay minerals such as illite or kaolinite, were more prominent in Shale 3, recording 0.64 wt% and 2.05 wt%, respectively. These values were comparatively lower in Shale 1 and Shale 2, reinforcing the presence of more clay structures in Shale 3, as also observed from FESEM analysis. The presence of clay contributes to matrix ductility, which may affect fracture propagation and sealing behavior.

Calcium (Ca) concentrations further confirmed the carbonate dominance in Shale 2 (40.67 wt%) and Shale 1 (29.46 wt%), whereas Shale 3 exhibited a much lower value of 10.65 wt%. The higher carbonate content in Shale 1 and 2 implies greater chemical reactivity when exposed to injected  $\text{CO}_2$ , potentially influencing dissolution or alteration processes. Iron (Fe) was detected in minor amounts across all samples, with Shale 3 showing the highest value (1.06 wt%), possibly due to the presence of hematite or iron-bearing clays.

Overall, the EDS results highlight significant compositional differences among the samples. Shale 1 and Shale 2 are carbonate-rich with lower quartz and clay contents, while Shale 3 is more siliceous and clay-bearing. These mineralogical variations are expected to influence both the geomechanical stability and  $\text{CO}_2$  rock interactions, with implications for long-term caprock sealing performance in CCS operations.

### X-ray fluorescence (XRF)

The X-ray fluorescence (XRF) analysis result in Table 1 revealed distinct mineralogical compositions across the three shale samples.

**Table 1—X-ray fluorescence (XRF) analysis showing the mineralogical composition of Shale 1, Shale 2, and Shale 3.**

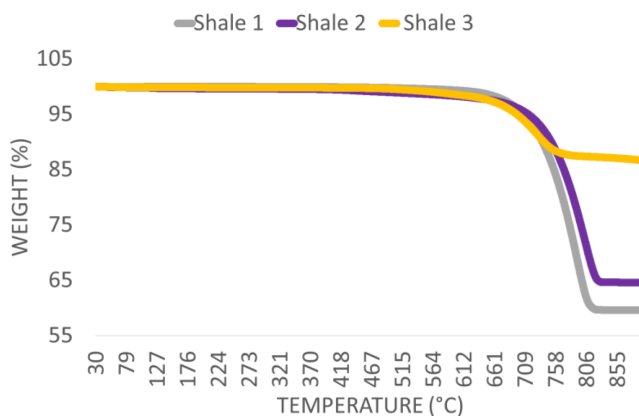
| Sample  | Shale (wt%) | Sandstone (wt%) | Clay (wt%) | Dominant Mineral Phases                                                                                                      |
|---------|-------------|-----------------|------------|------------------------------------------------------------------------------------------------------------------------------|
| Shale 1 | 99          | 0.5             | 0.1        | CaCO <sub>3</sub> (Carbonate), trace SiO <sub>2</sub> , trace Al <sub>2</sub> O <sub>3</sub>                                 |
| Shale 2 | 94          | 3               | 1          | CaCO <sub>3</sub> (Carbonate), SiO <sub>2</sub> , Al <sub>2</sub> O <sub>3</sub>                                             |
| Shale 3 | 53          | 20              | 2          | CaCO <sub>3</sub> (Carbonate), SiO <sub>2</sub> , Al <sub>2</sub> O <sub>3</sub> , Fe <sub>2</sub> O <sub>3</sub> (Hematite) |

Bulk mineralogical composition of the shales was analyzed using X-ray fluorescence (XRF). Shale 1 had a composition consisting of 99% shale, with small traces of sandstone (0.5%) and clay (0.1%). Likewise, Shale 2 is mainly composed of shale (94%), with slightly higher contents of sandstone (3%) and clay (1%). By comparison Shale 3 was found to have a significantly different composition of only 53% shale, containing markedly higher level of sandstone (20%) and slightly more clay (2%). These differences in composition indicate depositional environments of the samples.

The high content of shale in Shale 1 and Shale 2 reflects a more uniform fine-grained matrix, generally known for better sealing performance caused by low permeability. The occurrence of SiO<sub>2</sub> in quartz only was established in Sandstone and Al<sub>2</sub>O<sub>3</sub> in clay was determined in all samples; however, CaCO<sub>3</sub> characteristic of calcareous shale was a prominent component in Shale 1. The higher sandstone content in Shale 3 could diminish its effectiveness as a seal because coarser grain size and higher porosity are to be expected in most quartz-rich lithologies. Further, the higher hematite (Fe<sub>2</sub>O<sub>3</sub>) abundance in Shale 3 may imply oxidizing formation conditions that may affect geomechanical response to CO<sub>2</sub> injection. In general, the XRF data bring out the compositional diversity of the shales formations which controls the suitability and performances of the shales as caprocks in CCS projects.

### Thermogravimetric Analysis (TGA)

The thermal stability of Shale 1, Shale 2, and Shale 3 was evaluated using TGA under two different heating rates: 10 °C/min and 20 °C/min. At a heating rate of 10 °C/min (Fig. 3), all shale samples exhibited thermal decomposition onset at approximately 650 °C.

**Figure 3—Thermogravimetric analysis (TGA) of shale sample heated at a rate of 10 °C/min.**

Shale 3 showed the highest residual weight (~86%), indicating superior thermal stability compared to Shale 1 and Shale 2, which retained ~60% and ~65% of their weight, respectively. This higher stability suggests that Shale 3 possesses a more thermally resilient matrix, potentially due to its distinct mineralogical composition and lower organic matter content.

At an increased heating rate of 20 °C/min (Fig. 4), decomposition occurred earlier, beginning near 350 °C.

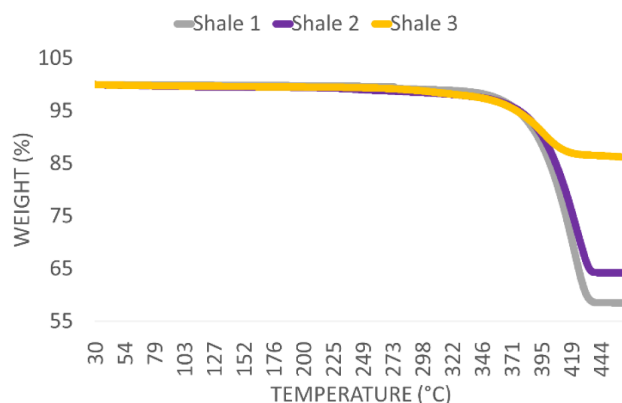


Figure 4—Thermogravimetric analysis (TGA) of shale sample heated at 20°C/min.

The acceleration of the decomposition rate with increased heating aligns with kinetic behavior expected in thermally sensitive geological samples. Despite the earlier decomposition, the trend in thermal stability remained consistent, with Shale 3 maintaining the highest residual weight and Shale 1 the lowest. This confirms that Shale 3 has superior resistance to thermal degradation under varying thermal stresses.

These results imply that Shale 3 may offer enhanced structural integrity under the thermal conditions associated with CO<sub>2</sub> injection and long-term storage, making it a potentially more secure caprock candidate for CCS applications. The reduced weight loss and delayed decomposition further support its suitability in high-temperature subsurface environments.

### Numerical Geomechanical Simulation

**In-Situ Stress and Injection.** For Shale 3 model, the CO<sub>2</sub> were injected at a constant rate of 6.86 MMscf/day for a consistent pressure buildup over time. During the early injection, the bottomhole pressure (BHP) peaked at 4,092 psi, then slowly stabilized between 3,060 and 3,300 psi for the rest of the timesteps. However, reservoir pressure increasing steadily from 2,684 psi to over 8,000 psi throughout the timesteps.

This simulated distribution of maximum stress in Z-direction shows how the stress responds to the constant gas injection from well INJ\_1. As shown in Fig. 5, the magnitude of stress ranges from -14.5 psi to 14.5 psi indicating a transition between compressive and tensile stress within the reservoir. The orange-red zone concentration indicates the tensile stress accumulation that may affect the caprock integrity. The compressive stress (blue-green zone) dominates the lower and lateral part, showing limited stress propagation across the formation.

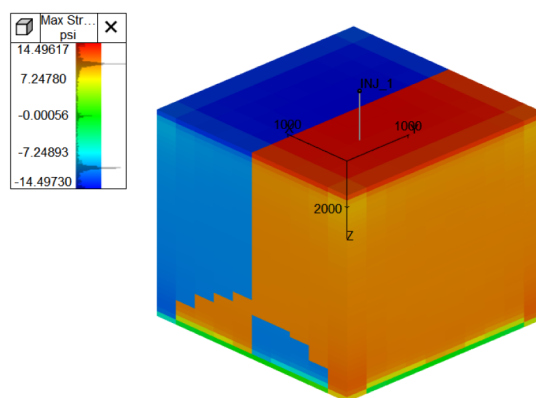


Figure 5—Distribution of Maximum Principal Stress  $\sigma_1$

Fig. 6 shows the distribution of minimum stress,  $\sigma_3$  in X-direction in the reservoir. The value of stress ranges from 0.001 psi to 14.5 psi, with the highest concentration located near top of the model around the injection well. This indicates that the upper part of reservoir is under more stress due to the pressure buildup from the injected gas.

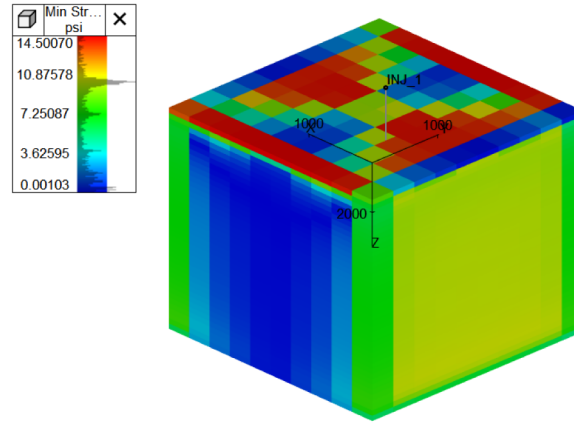


Figure 6—Distribution of Minimum Principal Stress  $\sigma_3$

**Mohr-Coulomb Failure Criterion.** The Mohr-Coulomb criterion was calculated to assess the potential of shear failure in the caprock. The Mohr-Coulomb criterion general equation expressed by

$$\tau = \sigma \tan(\theta) + c \quad (1)$$

where

$\tau$  is the shear stress

$\sigma$  is the normal stress

$\theta$  is the angle of internal friction

$c$  is the cohesion

The principal stress Mohr-Coulomb equation:

$$\sigma_1 - \sigma_3 \leq (\sigma_1 + \sigma_3) \sin \theta + 2c \cos \theta \quad (2)$$

where

$\sigma_1$  is the maximum principal stress

$\sigma_3$  is the minimum principal stress

$\theta$  is the angle of internal friction

$c$  is the cohesion

Based on the simulated stress output, the principal stress values were extracted around the injection well area. The cohesion and internal friction angle values used were 2175.57 psi and  $30^\circ$ , respectively. The results shows that the stress conditions remain well below the failure envelope throughout the simulation period. The calculated failure index is significantly lower than 1, confirming that the caprock remains mechanically stable under the current injection scenario. No indications of shear failure were observed in the stress regime surrounding the injection zone.

Similar studies on shale formations in the US showed shear failure due to  $\text{CO}_2$  injection because of the stress increases observed, especially near the injection well. In one instance, fractures were observed to initiate and propagate out toward the well bore under high pore pressure and in another, failure was localized near the injection zone but limited by certain stress orientations [12], [13]. In contrast, a more consistent stress distribution is presented by the current model, and no fracture initiation is observed and this may be the result of the differences in the initial stress state, material characteristics and the injection rate applied.

These comparisons emphasize the role of site-specific stress conditions in determining risk of failure during CO<sub>2</sub> injection into shales.

## Conclusion

This paper extensively investigates the geomechanical behavior of shale caprocks during CO<sub>2</sub> injection through the combination of mineralogical, morphological, thermal, mechanical, and simulation analyses. Shale from proven U.S. storage formations, and from a local formation, were assessed to determine whether they might be used for long-term carbon storage. All samples exhibited potential CO<sub>2</sub> capture, despite variations in their composition, with Shale 3 presenting stronger thermal stability and weaker mechanical character. On the other hand, Shale 1 and Shale 2 had higher fracture tolerance, however, with structural constraints suggested here. The results suggest that caprock integrity is controlled by a combination of geochemical and geomechanical processes. However, the results confirm that local shale formations like those studied, although different to their conventional counterparts, deliver adequate conditions for effective seal of CO<sub>2</sub>, provided that proper characterization is performed. This study, therefore, enables better decision making in safety and site selection in CCS operations through Malaysia and demonstrates that coupled laboratory testing and simulation software such as tNavigator can improve predictive capacity of caprock in CCS operations.

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